Tyrone Marhguy

Computer Engineering | Systems, Backend & Developer Tools
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Education

University of Pennsylvania — Philadelphia, PA
B.S.E. in Computer Engineering — Expected May 2028

Experience

Software Engineer Intern — May – Aug. 2026
Aragorn AI — Remote

- Contributed to **production backend and API-integration workflows** for AI services; debugged cross-service failures and documented interfaces for production integrations.

Open-Source Software Contributor — 2026 – Present
OpenROAD, Verilator, LibreLane, OpenFPGA — C++ / Python

- Fixed OpenROAD's LEF58 parser and Verilator's Linux peak-memory reporting; shipped Yosys compatibility and synthesis-reporting fixes in **LibreLane 3.0.8 and 3.0.10**.

Hardware & Firmware Engineer — May – Aug. 2026
Vero Electric — Philadelphia, PA

- Developed **embedded C/C++ fault-detection and SPI/I2C acquisition firmware** for a 16-channel battery monitor; debugged board integration on a **240 kWh / 120 kW** energy-storage platform.

Teaching Assistant | CIS 1100 — Aug. – Dec. 2025
University of Pennsylvania — Philadelphia, PA

- Authored **50+ pytest autograding suites for 300+ students**, covering edge cases, boundaries and regressions for weekly programming assignments.

Selected Software Projects

SeaLion | Distributed Full-Text Search Rust — Docs | Code

- Built Unicode analysis, inverted indexes, persistent segments and **BM25 ranking** from first principles; added exact Block-Max WAND, typo correction, crawling, and shards/replicas with failover.
- Validated search against a reference oracle; documented **155 passing tests** and release benchmarks of **5,114 docs/s indexing** and **271 queries/s** exact WAND search.

FigDB | Transactional Database Rust — Docs | Code

- Built a transactional database from first principles with a **checksummed WAL**, WAL-backed memtables, immutable SSTables, LSM flush/merge, manifests and compaction; implemented **crash recovery** and correctness tests.

Lobster | Systems Language & Compiler Rust — Docs | Code

- Built a lexer, **Pratt parser**, name resolution and type checker; lowered typed HIR/MIR into verified **SSA**, implemented optimization passes, and differential-tested execution with a reference interpreter.

FramePort | HDMI Capture in VS Code TypeScript / FFmpeg — Docs | Install

- Published a VS Code extension with **1,300+ Open VSX downloads** for live HDMI/USB capture, pixel inspection, PNG screenshots and **H.264 recording**; implemented backpressure, lifecycle recovery, safe saves and encode/decode regression coverage.

Envelop | Browser-to-FPGA Messaging TypeScript / Swift / Kotlin — Demo | Code

- Connected browser clients to physical Tomato computers through native BLE bridges; implemented a **durable job queue**, bridge leases, replay-safe handling and the binary **ENVELOP/1** protocol.

Technical Skills

Languages: Rust, C/C++, Python, TypeScript/JavaScript, Swift, Java/Kotlin, SQL, Bash
Systems: Linux, concurrency, networking, storage engines, search/indexing, compilers, binary protocols
Backend: FastAPI, Node.js, REST APIs, PostgreSQL, Supabase, Redis
Tooling: Docker, Kubernetes, GitHub Actions, CI/CD, Nginx, FFmpeg, VS Code API, pytest